

IT-968 Electrical Properties

High Speed, High Tg, Low Dk And Ultra Low Df Laminate & Prepreg

Core IT-968TC

Laminate			Dk						Df					
Imperial units Thickness (in)	Metric system Thickness (mm)	Standard Constructions	1 GHz	2 GHz	5 GHz	10 GHz	15 GHz	20 GHz	1 GHz	2 GHz	5 GHz	10 GHz	15 GHz	20 GHz
0.0020	0.051	1067x1	3.46	3.36	3.29	3.28	3.28	3.27	0.0031	0.0033	0.0037	0.0043	0.0046	0.0049
0.0020	0.051	1035x1	3.43	3.33	3.26	3.25	3.25	3.24	0.0031	0.0033	0.0037	0.0042	0.0045	0.0048
0.0020	0.051	106x1	3.34	3.24	3.18	3.17	3.17	3.15	0.0030	0.0032	0.0035	0.0040	0.0043	0.0045
0.0023	0.058	1067x1	3.36	3.26	3.20	3.19	3.19	3.17	0.0030	0.0032	0.0036	0.0040	0.0043	0.0046
0.0025	0.064	1078x1	3.70	3.60	3.53	3.52	3.52	3.50	0.0034	0.0036	0.0041	0.0049	0.0053	0.0057
0.0030	0.076	1086x1	3.61	3.51	3.44	3.43	3.42	3.41	0.0033	0.0035	0.0040	0.0046	0.0050	0.0054
0.0035	0.089	3313x1	3.97	3.87	3.78	3.77	3.77	3.76	0.0036	0.0038	0.0045	0.0055	0.0060	0.0064
0.0040	0.102	2116x1	4.23	4.13	4.03	4.03	4.03	4.02	0.0038	0.0040	0.0049	0.0061	0.0067	0.0072
0.0040	0.102	3313x1	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0040	0.102	1035X2	3.43	3.33	3.26	3.25	3.25	3.24	0.0031	0.0033	0.0037	0.0042	0.0045	0.0048
0.0040	0.102	106x2	3.34	3.24	3.18	3.17	3.17	3.15	0.0030	0.0032	0.0035	0.0040	0.0043	0.0045
0.0045	0.114	2116x1	4.00	3.90	3.81	3.81	3.81	3.80	0.0036	0.0038	0.0046	0.0056	0.0061	0.0065
0.0045	0.114	1067x2	3.36	3.26	3.20	3.19	3.19	3.17	0.0030	0.0032	0.0036	0.0040	0.0043	0.0046
0.0050	0.127	2116x1	3.84	3.74	3.65	3.65	3.65	3.63	0.0035	0.0037	0.0044	0.0052	0.0057	0.0061
0.0050	0.127	1080x2	3.70	3.60	3.53	3.52	3.52	3.50	0.0034	0.0036	0.0041	0.0049	0.0053	0.0057
0.0050	0.127	1078X2	3.70	3.60	3.53	3.52	3.52	3.50	0.0034	0.0036	0.0041	0.0049	0.0053	0.0057
0.0057	0.145	1078X2	3.55	3.45	3.38	3.37	3.37	3.36	0.0032	0.0034	0.0039	0.0045	0.0049	0.0052
0.0060	0.152	1086x2	3.61	3.51	3.44	3.43	3.42	3.41	0.0033	0.0035	0.0040	0.0046	0.0050	0.0054
0.0070	0.178	3313x2	3.97	3.87	3.78	3.77	3.77	3.76	0.0036	0.0038	0.0045	0.0055	0.0060	0.0064
0.0080	0.203	2116x2	4.23	4.13	4.03	4.03	4.03	4.02	0.0038	0.0040	0.0049	0.0061	0.0067	0.0072
0.0080	0.203	3313x2	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0100	0.254	2116x2	3.84	3.74	3.65	3.65	3.65	3.63	0.0035	0.0037	0.0044	0.0052	0.0057	0.0061
0.0120	0.305	2116x3	4.23	4.13	4.03	4.03	4.03	4.02	0.0038	0.0040	0.0049	0.0061	0.0067	0.0072
0.0120	0.305	3313x3	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0150	0.381	2116x3	3.84	3.74	3.65	3.65	3.65	3.63	0.0035	0.0037	0.0044	0.0052	0.0057	0.0061
0.0150	0.381	3313x4	3.86	3.76	3.67	3.67	3.67	3.65	0.0035	0.0037	0.0044	0.0052	0.0057	0.0061
0.0160	0.406	2116x4	4.23	4.13	4.03	4.03	4.03	4.02	0.0038	0.0040	0.0049	0.0061	0.0067	0.0072
0.0160	0.406	3313x4	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0180	0.457	2116x4	4.00	3.90	3.81	3.81	3.81	3.80	0.0036	0.0038	0.0046	0.0056	0.0061	0.0065
0.0180	0.457	3313x4	3.62	3.52	3.45	3.44	3.44	3.43	0.0033	0.0035	0.0040	0.0047	0.0051	0.0054
0.0200	0.508	2116x4	3.84	3.74	3.65	3.65	3.65	3.63	0.0035	0.0037	0.0044	0.0052	0.0057	0.0061
0.0200	0.508	3313x5	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0210	0.533	2116x5	4.13	4.03	3.93	3.93	3.93	3.92	0.0037	0.0039	0.0048	0.0059	0.0065	0.0069
0.0210	0.533	3313x5	3.70	3.60	3.53	3.52	3.52	3.50	0.0034	0.0036	0.0041	0.0049	0.0053	0.0057
0.0240	0.610	2116x5	3.90	3.80	3.71	3.71	3.71	3.69	0.0035	0.0037	0.0044	0.0053	0.0059	0.0062
0.0240	0.610	3313x6	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0280	0.711	2116x6	3.94	3.84	3.75	3.75	3.75	3.74	0.0036	0.0038	0.0045	0.0054	0.0060	0.0064
0.0280	0.711	3313x7	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0300	0.762	2116x6	4.09	3.99	3.89	3.89	3.89	3.88	0.0037	0.0039	0.0047	0.0058	0.0064	0.0068
0.0320	0.813	2116x8	4.23	4.13	4.03	4.03	4.03	4.02	0.0038	0.0040	0.0049	0.0061	0.0067	0.0072
0.0320	0.813	3313x8	3.77	3.67	3.59	3.58	3.58	3.57	0.0034	0.0036	0.0042	0.0050	0.0055	0.0059
0.0340	0.864	2116x8	4.13	4.03	3.93	3.93	3.93	3.92	0.0037	0.0039	0.0048	0.0059	0.0065	0.0069
0.0380	0.965	2116x9	4.13	4.03	3.93	3.93	3.93	3.92	0.0037	0.0039	0.0048	0.0059	0.0065	0.0069

Prepreg IT-968BS

Laminated Prepreg			Dk						Df					
Prepreg Type	Thickness (mil)	Resin Content (%)	1 GHz	2 GHz	5 GHz	10 GHz	15 GHz	20 GHz	1 GHz	2 GHz	5 GHz	10 GHz	15 GHz	20 GHz
1037	2.0	73.0	3.24	3.14	3.08	3.07	3.06	3.05	0.0029	0.0031	0.0033	0.0037	0.0040	0.0042
	2.3	75.0	3.19	3.09	3.04	3.02	3.02	3.00	0.0028	0.0030	0.0032	0.0035	0.0038	0.0040
106	2.1	71.5	3.28	3.18	3.12	3.10	3.10	3.09	0.0029	0.0031	0.0034	0.0038	0.0041	0.0043
	2.3	73.0	3.24	3.14	3.08	3.07	3.06	3.05	0.0029	0.0031	0.0033	0.0037	0.0040	0.0042
	2.5	75.0	3.19	3.09	3.04	3.02	3.02	3.00	0.0028	0.0030	0.0032	0.0035	0.0038	0.0040
1035	2.3	72.5	3.25	3.15	3.09	3.08	3.08	3.06	0.0029	0.0031	0.0034	0.0037	0.0040	0.0042
	2.5	74.0	3.21	3.11	3.06	3.04	3.04	3.03	0.0028	0.0030	0.0033	0.0036	0.0039	0.0041
	2.7	76.0	3.16	3.06	3.01	2.99	2.99	2.98	0.0028	0.0030	0.0032	0.0035	0.0037	0.0039
1067	2.3	71.5	3.28	3.18	3.12	3.10	3.10	3.09	0.0029	0.0031	0.0034	0.0038	0.0041	0.0043
	2.5	73.0	3.24	3.14	3.08	3.07	3.06	3.05	0.0029	0.0031	0.0033	0.0037	0.0040	0.0042
	2.7	75.0	3.19	3.09	3.04	3.02	3.02	3.00	0.0028	0.0030	0.0032	0.0035	0.0038	0.0040
1078	2.8	62.0	3.55	3.45	3.38	3.37	3.37	3.36	0.0032	0.0034	0.0039	0.0045	0.0049	0.0052
	3.1	65.0	3.46	3.36	3.29	3.28	3.28	3.27	0.0031	0.0033	0.0037	0.0043	0.0046	0.0049
	3.5	68.0	3.37	3.27	3.21	3.19	3.19	3.18	0.0030	0.0032	0.0036	0.0040	0.0044	0.0046
	3.8	70.0	3.32	3.22	3.16	3.14	3.14	3.13	0.0029	0.0031	0.0035	0.0039	0.0042	0.0044
1080	2.8	62.0	3.55	3.45	3.38	3.37	3.37	3.36	0.0032	0.0034	0.0039	0.0045	0.0049	0.0052
	3.1	65.0	3.46	3.36	3.29	3.28	3.28	3.27	0.0031	0.0033	0.0037	0.0043	0.0046	0.0049
	3.5	68.0	3.37	3.27	3.21	3.19	3.19	3.18	0.0030	0.0032	0.0036	0.0040	0.0044	0.0046
1086	3.0	61.0	3.59	3.49	3.41	3.40	3.40	3.39	0.0032	0.0034	0.0040	0.0046	0.0050	0.0053
	3.5	65.0	3.46	3.36	3.29	3.28	3.28	3.27	0.0031	0.0033	0.0037	0.0043	0.0046	0.0049
	3.9	68.0	3.37	3.27	3.21	3.19	3.19	3.18	0.0030	0.0032	0.0036	0.0040	0.0044	0.0046
2113	3.9	56.0	3.76	3.66	3.58	3.57	3.57	3.56	0.0034	0.0036	0.0042	0.0050	0.0055	0.0058
	4.1	58.0	3.69	3.59	3.51	3.50	3.50	3.49	0.0033	0.0035	0.0041	0.0048	0.0053	0.0056
	4.4	60.0	3.62	3.52	3.45	3.44	3.43	3.42	0.0033	0.0035	0.0040	0.0047	0.0051	0.0054
3313	4.0	56.0	3.76	3.66	3.58	3.57	3.57	3.56	0.0034	0.0036	0.0042	0.0050	0.0055	0.0058
	4.3	58.0	3.69	3.59	3.51	3.50	3.50	3.49	0.0033	0.0035	0.0041	0.0048	0.0053	0.0056
	4.6	60.0	3.62	3.52	3.45	3.44	3.43	3.42	0.0033	0.0035	0.0040	0.0047	0.0051	0.0054
2116	4.8	53.0	3.88	3.78	3.70	3.69	3.69	3.68	0.0035	0.0037	0.0044	0.0053	0.0058	0.0062
	5.1	55.0	3.80	3.70	3.62	3.61	3.61	3.60	0.0034	0.0036	0.0043	0.0051	0.0056	0.0060
	5.4	57.0	3.73	3.63	3.55	3.54	3.54	3.53	0.0034	0.0036	0.0042	0.0049	0.0054	0.0057

Test methods: 1GHz by RF impedance/material analyzer HP4291B (IPC 2.5.5.9);
2GHz, 5GHz, 10GHz,15GHz, 20GHz by Resonater Cavity (IPC 2.5.5.13)

Note: Dk value should be subtracted 0.3 to be target data for impedance caculation

The above data and fabrication guide provide designers and PCB shop for their reference. We truly believe that these information are accurate, moreover, test methods and specification must be consider interiorly. The effective sale is in accordance with the specification, and ITEQ maintains the authority of revision to keep the best suitability for users.